

**TUTORLINKAI: A Verified Local Tutor
Discovery and Booking Platform**

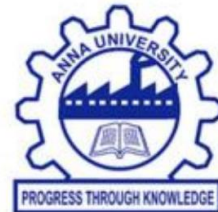
PHASE I REPORT

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*in partial fulfillment for the award of the degree
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BONAFIDE CERTIFICATE

Certified that this Report titled **“TUTORLINKAI: A VERIFIED LOCAL TUTOR DISCOVERY AND BOOKING PLATFORM”** is the bonafide work of **AWINTHIKA SANTHANAM (230701048), ABHINAYA LAKSHMI S (230701004), ABIRAMI K (230701008)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

“This project addresses the following Sustainable Development Goal(s): **Goal 4: Quality Education; Goal 10: Reduced Inequalities.**”

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ABSTRACT

Students who find it difficult to keep pace with classroom instruction often need additional academic support, and parents seek tutors who can help their children with the concepts in which they need more assistance. Tutors are commonly chosen because they live close by or have been suggested by relatives or neighbours, without background checks or an assessment of whether the tutor is qualified and appropriate for the needs of the student. The lack of tutor verification, together with the absence of structured means of finding and comparing tutors based on individual learning requirements, is a major problem faced by parents and students.

This project presents TutorLinkAI, a secure and intelligent local tutor discovery and booking platform that connects students and parents with verified tutors based on individual learning requirements. The proposed system consists of authentication and user management, student learning need management, tutor onboarding and profile management, tutor verification, AI-based tutor matching, explainable recommendation and booking modules. During onboarding, tutors register with personal, academic and professional information and upload qualification documents, which are processed using Optical Character Recognition (OCR). The extracted information is compared with the registered profile using similarity-based validation, while document security analysis, facial analysis and liveness assessment provide additional verification evidence. Cases requiring further assessment can be handled through a manual verification review process.

Students can create and manage multiple learning needs containing subjects, specific topics, learning goals, academic level, preferred tutor characteristics, budget, availability and location. The matching system represents student requirements and tutor expertise using sentence embeddings and combines semantic similarity with topic compatibility and structured contextual factors such as location, fee and availability. The resulting tutors are ranked based on their compatibility with the selected learning need, with supporting match information provided to improve the explainability of recommendations. The platform is implemented using a React-based frontend, a Python and FastAPI backend, a PostgreSQL database with SQLAlchemy and Alembic, and dedicated AI/ML components.

Overall, TutorLinkAI integrates tutor credential verification, learning-need-aware tutor matching, explainable recommendations and booking into a unified platform. The system provides a structured approach to identifying suitable tutors by considering both tutor credibility and the individual academic requirements of the student.

Keywords: TutorLinkAI, Tutor Discovery, Tutor Verification, Credential Verification, Optical Character Recognition (OCR), Semantic Matching, Explainable Recommendation, Artificial Intelligence, Educational Technology, Tutor Recommendation, Academic Support

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
KNN	K-Nearest Neighbours
LBS	Location-Based Service
NLP	Natural Language Processing
OCR	Optical Character Recognition
REST	Representational State Transfer
SDG	Sustainable Development Goal
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Learning has increasingly moved beyond the confines of a traditional classroom. Given the academic background of students and their level of comprehension, classroom teaching may not always be sufficient, and tuition services have long provided additional support to students who require more help in certain areas of learning than what is provided in the classroom. However, the efficiency of such assistance depends on locating a tutor who is qualified, experienced, and able to guide the student according to specific needs.

Traditionally, students and parents have relied on private networks such as family, friends and neighbours to locate tutors. While such methods can be effective in shortlisting potential tutors, the information provided is usually limited. Factors to consider when selecting a tutor include qualifications, the subject matter or area of expertise in which the student needs help, experience, location, availability and fees. Gathering and comparing this information is a tedious process which many parents and students find inconvenient.

Online services have been developed to enable users to research and locate a suitable service based on certain criteria. However, there are various limitations associated with existing tutor discovery methods and services. Many existing platforms do not allow parents or students to easily locate tutors using specific criteria or to compare tutors against specific requirements. In many cases, finding a tutor relies heavily on word of mouth, which restricts students and parents to tutors within their personal networks and provides limited opportunities to match tutors effectively with the requirements of the student.

Parents and students must also engage in separate communication channels when contacting a tutor. They must individually contact a tutor to confirm qualifications and availability and to schedule and reschedule sessions, which is inconvenient and in many cases redundant. An equally important limitation is the reliability of the information provided. Educational background, experience and area of expertise are vital factors in selecting a tutor,

and making such information easily accessible helps both parties to hold realistic expectations of the tutor's qualifications and ability.

1.2 OBJECTIVE

The main objective of the proposed system, TutorLinkAI, is to provide parents and students with reliable information about tutors so that the right academic support can be obtained without adding strain on the student or the family. The specific objectives of the project are as follows:

1. To enable tutor discovery based on subject, location, qualifications, availability, experience and affordability, thereby reducing dependence on informal tutor recommendations.
2. To verify tutor credentials using Optical Character Recognition (OCR), profile–credential consistency checking, document security analysis, facial analysis and liveness assessment, with manual review support for cases requiring additional verification.
3. To allow students to create and manage multiple learning needs containing subjects, specific topics, areas of difficulty, learning objectives, academic level, preferred teaching characteristics, budget, availability and location.
4. To match students with suitable verified tutors using semantic similarity, topic compatibility and structured contextual factors, and to provide ranked and explainable tutor recommendations.
5. To provide a single platform through which parents, students and tutors can review profiles, communicate and manage session bookings.
6. To support United Nations Sustainable Development Goal (SDG) 4: Quality Education and SDG 10: Reduced Inequalities.

1.3 EXISTING SYSTEM AND PROBLEM IDENTIFICATION

Existing solutions are mostly traditional coaching institutes whose main goal is to increase student enrolment every year. This causes a financial drain on families and prevents students from finding the right help. These institutes often conduct classes for 20–30 students

at once, leading to a lack of personalisation and a failure to cater to the individual needs of each student. Most centres also assign homework to the students, adding to their physical and mental exhaustion, which in turn affects academic performance. It is nevertheless important to recognise the few centres that genuinely upskill students and help them improve their academic capabilities through extensive coaching.

Some existing online tutor finder platforms map students to tutors who reside in a different pincode, which adds to the physical exhaustion of the students. Such platforms often advertise elaborate courses that end up as clickbait for increased enrolments, and choosing from a wide variety of courses and faculties can become hectic. Students usually have limited information about the availability of tutors in their localities, about tutors who can help in the subjects in which they are struggling, and about tutors whose fees are affordable for them.

1.3.1 COMMERCIAL COACHING INSTITUTES AND MANUAL TUITION CENTRES

Students usually find local tuition centres or coaching classes through advertisements, recommendations or by asking around. The classes can be expensive or conducted in groups, leaving the tutor unable to cater to individual student needs. Finding the right tutor also takes time because parents often have to make calls or visit different centres.

1.3.2 WEB AND DIRECTORY-BASED TUTOR PORTALS

Examples of such portals include A+ Home Tuition and bdtutors.com. These websites let students search for tutors using basic details such as subject and location and do not focus on important factors such as student needs, affordability and class scheduling. They do not provide direct communication between tutors and parents, so arranging classes may take extra time. The matching is mostly based on profile details rather than on what the student actually needs help with.

1.3.3 MULTI-CRITERIA AND LOCATION-BASED PLATFORMS

Examples include the GRAB Tutor Management System (GTMS), Tutors Map and Tutor Finder (LBS). These platforms allow students to search for tutors using filters based on subject, gender and expertise, and they mainly use the information given in the tutor's profile.

They do not consider factors such as the student’s weak areas, teaching quality, availability and fees.

Overall, most existing solutions focus on finding tutors using basic information or past activity. They do not give sufficient importance to the student’s actual learning difficulties, current performance, tutor availability and cost. Moreover, there is no reliable verification of the tutor’s identity and qualifications, so parents cannot assess whether a tutor is legitimate and properly qualified. These gaps form the problem addressed by this project.

1.4 PROPOSED SYSTEM

The proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at addressing the limitations of traditional methods of tutor discovery. By aggregating tutor-related information and offering a range of features for both parents and tutors, TutorLinkAI facilitates the tutor discovery process based on subject, location, qualifications, availability, experience and affordability. Tutors can post their information and experience for students and parents to view, which addresses the limitation of relying on word of mouth by offering a wider pool of tutors. Students and parents can also sort through the available tutors according to their specific needs, eliminating the need to manually sift through numerous tutors or advertisements.

A key feature of the system is tutor credential verification. Tutors upload qualification documents during onboarding; the documents are processed using OCR, and the extracted information is compared with the registered profile to identify inconsistencies and to assign a verification status. The application therefore combines identity onboarding, credential verification and tutor suitability assessment within one platform, providing a structured alternative to relying solely on recommendations from relatives, neighbours or nearby tutors.

The proposed matching component goes beyond subject-level filtering by modelling the student's learning requirements at topic and goal level. Students can define multiple learning needs containing specific areas of difficulty, learning objectives, academic level, preferred teaching characteristics, budget, availability and location. Student requirements and tutor expertise are represented using semantic embeddings, while topic compatibility and contextual factors are incorporated into the ranking process. The resulting recommendations

provide supporting match information, such as matched topics and relevant compatibility factors, allowing students to understand the basis of each recommendation.

The proposed platform supports SDG 4: Quality Education by enabling students to receive more personalised supplementary education, through proper matching of students and tutors based on student requirements as well as tutor qualifications. It also aligns with SDG 10: Reduced Inequalities, since providing an affordable means of accessing supplementary education and the option of sorting tutors by budget can help reduce disparities in access to additional academic services.

CHAPTER 2

LITERATURE SURVEY

This chapter provides a theoretical framework and identifies prior research relevant to the design and development of intelligent online tutor discovery and matching services. Existing solutions primarily aim to bridge the gap between students and suitable tutors through subject-based searches, location-based discovery and booking facilities.

2.1 LITERATURE REVIEW

The early approaches to tutor discovery depend predominantly on the profile information provided by the user and on predefined search categories, providing limited support for personalised and student-centric tutor recommendations [6]–[8]. Several online tutor discovery systems have investigated structured student–tutor management and filtering mechanisms, incorporating criteria such as education level, subject expertise and geographical location for tutor filtering. In a comparative analysis of existing applications, student and tutor registration and tutor filtering are common functionalities, while features such as tutor–parent interaction and communication and tutor ranking provide additional support. These approaches demonstrate the feasibility of recommendation-based tutor discovery while indicating the importance of incorporating multiple learner and tutor characteristics [2].

Location-based approaches have improved tutor discovery by incorporating geographical context such as latitude and longitude information to map tutor and student locations and to filter tutors by proximity. This is particularly practical where travelling distance is a priority. Beyond online tutor-finding applications, location-based service recommendation systems combine proximity with other attributes. These systems employ the Haversine formula to calculate geographical distance and use K-Nearest Neighbours (KNN) with scoring based on factors such as proximity, price and ratings to rank service providers, while supporting provider availability, booking and reviews. Such approaches demonstrate that geographical proximity can be combined with other factors to determine the most suitable service provider. However, proximity alone is not sufficient to determine academic suitability and needs to be combined with features such as subject expertise, tutor reviews and teaching approach.

The limitations of static filtering have led to personalised recommendation systems that incorporate user preferences, behavioural information and service provider characteristics to generate relevant recommendations [6], [14]. Machine-learning-based recommendation systems use historical data such as user-provider relationships and combine it with contextual re-ranking and factors such as urgency, provider availability, ratings and user interactions, moving beyond simple filtering by learning user-provider relationships. For tutor recommendation, the same principle can be applied by treating students and parents as users and tutors as education providers, with additional features such as subject and topic expertise, academic level, student skill requirements, teaching preferences and learning objectives.

An important extension to conventional tutor recommendation is the incorporation of student skill and knowledge assessment. A student may identify a subject as a requirement without being able to identify the specific concepts in which additional assistance is required. Knowledge-tracing techniques address this by estimating a learner's evolving knowledge state from historical learning interactions [1]. One such approach is a Transformer-based Deep Knowledge Tracing framework for formative assessment of student skills [9], which analyses sequences of student interactions to estimate knowledge states and identify knowledge gaps. The resulting learner representation can distinguish between a student needing assistance across an entire subject and a student who requires help with specific topics, and it can therefore serve as an input for identifying suitable tutors.

Semantic representation of textual information provides another means of improving recommendation beyond keyword matching. Sentence embedding models represent textual information in vector spaces in which semantic relationships are measured computationally. Ni et al. proposed Sentence-T5 for scalable sentence encoding and evaluated its effectiveness for semantic textual similarity tasks [10]. This provides a basis for comparing descriptive student requirements with tutor expertise, particularly when different terms are used to express related concepts.

Automated document analysis can further support credential verification by identifying potentially manipulated documents. Al-Ameri et al. investigated an unsupervised method for detecting document forgery using document analysis techniques [3]. Such approaches show that document-level analysis can complement text extraction and profile consistency checks.

In the current TutorLinkAI implementation, OCR is primarily used to extract information from the submitted certificates, while advanced document forgery detection is a potential future extension.

The reviewed research areas provide the individual components that inform TutorLinkAI. Semantic representation supports comparison between descriptive student requirements and tutor expertise, while multi-criteria recommendation provides a basis for incorporating contextual factors such as location, pricing and availability. Credential verification research motivates the use of automated document processing and identity-related checks. TutorLinkAI integrates these aspects into a single workflow by combining tutor verification with learning-need-aware tutor recommendation.

2.2 COMPARISON AND DISCUSSION

The literature indicates that tutor discovery can be viewed as a multi-criteria recommendation problem. Educational compatibility can be determined from factors such as student skill requirements, subject or topic preferences, and tutor skills and expertise, while contextual feasibility can be incorporated through factors such as pricing, availability, distance, ratings and tutor profile verification. Table 2.1 summarises the approaches reviewed and the aspect of the problem addressed by each.

Table 2.1. Summary of Approaches Reviewed in the Literature

S.No.	Approach	Aspect Addressed	Ref.
1	Profile-based tutor discovery and filtering	Filtering by education level, subject expertise and location; registration, tutor–parent interaction and tutor ranking	[2], [6]–[8]
2	Location-based recommendation	Proximity using geographical coordinates; Haversine distance and KNN-based scoring with price and ratings	—
3	Machine-learning-based personalised recommendation	User preferences, behavioural information and contextual re-ranking	[6], [14]
4	Knowledge tracing	Estimation of a learner’s knowledge state and identification of knowledge gaps	[1], [9]
5	Sentence embeddings	Semantic comparison of student requirements and tutor expertise	[10]
6	Document forgery detection	Unsupervised detection of manipulated documents	[3]

Existing tutor recommendation systems demonstrate the importance of profile-based matching; location-based systems demonstrate the usefulness of geographical and service constraints; machine-learning-based recommendation systems highlight the importance of behavioural and contextual personalisation; and knowledge-tracing research highlights the importance of student assessment for identifying knowledge levels and mapping student needs. However, these capabilities are generally addressed as separate problems rather than being integrated into a unified tutor recommendation process. TutorLinkAI attempts to bring tutor verification, student requirement profiling and semantic matching together within a single platform.

2.3 CONCLUSION

The survey shows that existing approaches address individual aspects of tutor discovery, namely profile-based filtering, proximity, personalised recommendation, learner assessment, semantic similarity and document analysis, but do not provide an integrated process that also verifies the credibility of the tutor. This observation motivates the design of

TutorLinkAI as a modular platform that combines tutor onboarding, credential verification, student and tutor profiling, AI-based matching and booking.

CHAPTER 3

SYSTEM DESIGN

The proposed TutorLinkAI system is designed as a modular intelligent platform for tutor discovery and personalised tutor recommendation. The system integrates student and tutor profiling, tutor verification, student requirement assessment, AI-based matching and booking within a unified platform. The architecture separates the user interface, application logic, data management and AI-based processing components to support modular development and future extension.

3.1 SYSTEM ARCHITECTURE

TutorLinkAI follows a modular client–server architecture consisting of a React-based frontend, a FastAPI backend, a PostgreSQL database and dedicated AI/ML components. The frontend provides interfaces for student and tutor registration, profile management, learning need management, tutor discovery, recommendation and booking, along with a reviewer interface for verification cases requiring manual assessment. The backend manages authentication, user and profile management, learning needs, verification workflows, matching requests, recommendation data and booking operations. Dedicated AI/ML components support semantic representation and tutor matching, while document processing and facial analysis services support credential verification. PostgreSQL provides persistent storage for user profiles, learning needs, tutor expertise, verification records, bookings and related application data. The overall architecture of the system is shown in Figure 3.1.

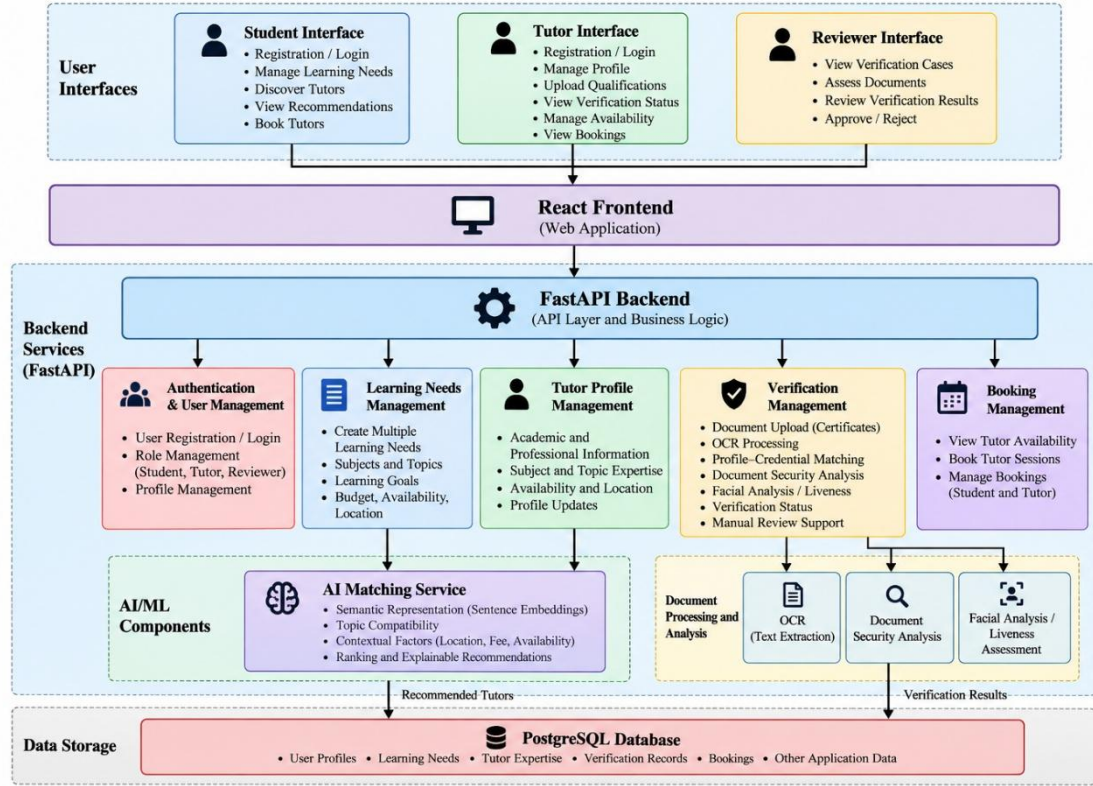


Figure 3.1. Overall System Architecture of TutorLinkAI

As shown in Figure 3.1, the architecture is organised into presentation, backend service, AI/ML processing and data storage layers. The presentation layer, built with React, provides role-specific interfaces for students, tutors and verification reviewers. The backend layer, built with FastAPI, manages authentication, user and profile management, learning needs, verification workflows, matching and recommendation, and booking operations. The AI/ML layer supports semantic representation and tutor matching, while document processing and facial analysis components support credential verification. The database layer, built on PostgreSQL, stores user profiles, learning needs, tutor expertise, verification records, availability, recommendations, bookings and related application data.

3.1.1 STUDENT LEARNING NEED WORKFLOW

The student learning need workflow allows students to represent different academic requirements as separate learning needs. Each learning need can contain subjects, specific topics, learning goals, academic level, preferred tutor characteristics, budget, availability and location. Students can create, edit, activate or remove learning needs according to their current

academic requirements. The selected active learning need is used as an input to the tutor matching and recommendation process.

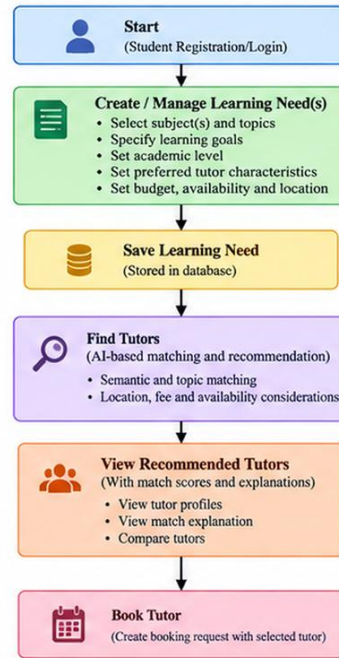


Figure 3.2. Student Learning Need Workflow

The workflow allows the recommendation process to be targeted to a specific academic requirement rather than treating all student requirements as a single static profile. This enables the same student to maintain different learning requirements and receive recommendations according to the learning need currently selected.

3.1.2 TUTOR ONBOARDING AND VERIFICATION WORKFLOW

The tutor onboarding and verification workflow separates account registration from qualification verification. After registering and completing the tutor profile, the tutor submits qualification documents for automated verification. The verification pipeline extracts information from the submitted documents, performs profile and credential consistency checks, analyses document security and evaluates available facial and liveness evidence. Cases requiring additional assessment can be routed to a manual verification reviewer. Tutors that satisfy the required verification conditions become eligible for tutor discovery and booking.

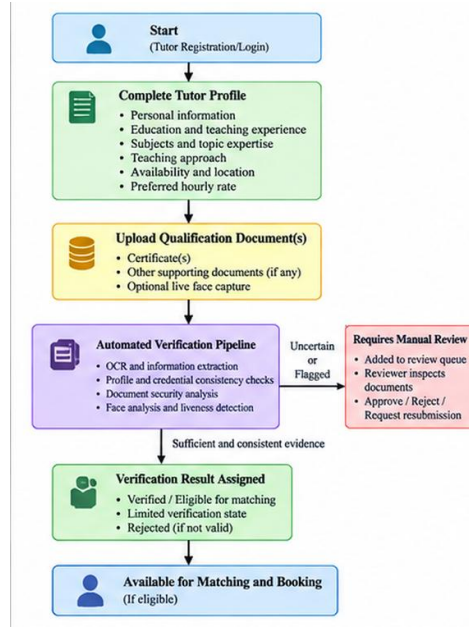


Figure 3. 3. Tutor Onboarding and Verification Workflow

The separation between onboarding and verification ensures that registration alone does not establish tutor credibility. Qualification information and identity-related evidence are assessed before the tutor is made eligible for production recommendations.

3.2 SYSTEM REQUIREMENTS

The software technologies used in the implementation and design of TutorLinkAI are identified in the following section.

3.2.1 SOFTWARE REQUIREMENTS

The software components used by the proposed system are listed in Table 3.1.

Table 3.1. Software Requirements of the Proposed System

S.No.	Component	Technology	Purpose
1	Frontend	React	Interfaces for registration, profile management, assessment, tutor discovery, recommendation and booking
2	Backend	Python, FastAPI	RESTful APIs for authentication, onboarding, profile management, document processing and verification
3	Database	PostgreSQL	Persistent data storage
4	Database interaction and migration	SQLAlchemy, Alembic	Database interaction and database schema migration
5	Image processing and face verification	OpenCV, SFace	Face detection, facial representation and identity similarity analysis
6	Certificate OCR	PaddleOCR	Extraction of name, institution, qualification and other textual fields from certificates
7	Semantic matching	Sentence Transformers	Sentence embeddings of student requirements and tutor expertise

3.3 END-TO-END WORKFLOW

The system provides an end-to-end workflow covering tutor onboarding, credential verification, student learning need modelling, tutor matching, recommendation and booking. Tutors register and complete their profiles before submitting qualification documents for verification. The verification pipeline extracts certificate information using OCR, checks consistency with the registered profile, performs document security analysis and evaluates available facial and liveness evidence. Cases requiring further assessment can be routed to a manual verification reviewer. A certificate without a usable portrait is processed using the remaining available verification modalities rather than being rejected solely due to the absence of a facial modality.

Students register and create one or more learning needs describing their academic requirements, topics, learning goals and contextual preferences. The matching component compares student learning needs with verified tutor expertise using semantic similarity, topic compatibility and contextual factors including location, fee and availability. Ranked tutor

recommendations are presented with supporting match information. The selected tutor can then be booked through the booking workflow, which maintains the relationship between the student, tutor and selected learning need.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

TutorLinkAI TutorLinkAI is a secure and intelligent platform designed to support students in discovering credible and suitable tutors based on their academic requirements. The system addresses two major challenges in tutor selection: the difficulty of assessing tutor suitability and the lack of reliable tutor verification. The proposed methodology integrates user authentication, student learning need management, tutor onboarding and profiling, multimodal credential verification, AI-based tutor matching, explainable recommendation and booking into a unified workflow.

During onboarding, students and tutors register on the platform through role-specific workflows. Students can create and manage one or more learning needs containing information such as subjects, specific topics, learning goals, academic level, preferred tutor characteristics, budget, availability and location. Tutors create profiles containing their personal, academic and professional information, including qualifications, areas of expertise, teaching preferences, availability, location and pricing. Registration and qualification verification are treated as separate stages so that the creation of a tutor account does not by itself establish tutor credibility.

Tutor qualification documents are processed through a multimodal verification workflow. Uploaded certificates are analysed using PaddleOCR to extract relevant information such as the tutor's name, institution, qualification and other available fields. The extracted information is compared with the registered tutor profile to identify inconsistencies, while document security analysis is performed to identify potential document-level risks. When a usable portrait is available, facial evidence can be evaluated using OpenCV-based face detection and SFace-based facial representation. Live-face frames and image-based liveness analysis provide additional identity-related evidence. Verification cases that require further assessment can be routed to a manual verification reviewer before the tutor becomes eligible for production recommendations.

The AI-based matching component compares a selected student learning need with verified tutor profiles using semantic similarity, topic compatibility and contextual factors. Sentence embeddings are used to represent student requirements and tutor expertise so that semantically related descriptions can be matched even when different terminology is used. The ranking process also considers geographical proximity, fee compatibility and availability, with teaching mode, language and pedagogical compatibility incorporated when the corresponding information is available. Ranked recommendations are presented with supporting match information to improve the interpretability of the recommendation process.

The backend of the application is implemented using Python and FastAPI, providing RESTful services for authentication, onboarding, learning need management, verification, matching, recommendation and booking. PostgreSQL is used for persistent data storage, with SQLAlchemy providing database interaction and Alembic supporting database schema migration. The React-based frontend provides role-specific interfaces for students, tutors and verification reviewers. Once a tutor satisfies the required verification conditions, the tutor becomes eligible for discovery and booking, providing an end-to-end workflow from student learning need creation and tutor verification to AI-based recommendation and booking.

4.2 MODULES

The system is composed of the following modules, which are described in this section.

4.2.1 AUTHENTICATION AND USER MANAGEMENT MODULE

This module manages user registration, authentication and role-based access within TutorLinkAI. Separate access paths are provided for students, tutors and verification reviewers. User authentication is handled through the backend authentication service, while protected API endpoints enforce access based on the authenticated user's role and ownership of the requested resources.

The module provides the foundation for protecting student learning information, tutor profile data, qualification documents, verification records and booking information. Role-based access ensures that students can manage their own learning requirements and bookings, tutors can manage their profiles and verification submissions, and authorized verification reviewers can access cases requiring manual assessment.

4.2.2 STUDENT LEARNING NEED MANAGEMENT MODULE

This module allows students to represent and manage their academic requirements in a structured manner. Instead of restricting a student to a single static requirement profile, the system supports the creation and management of multiple learning needs. Each learning need represents a specific academic requirement that can be used independently during the tutor matching process.

A learning need can contain information such as subjects, specific topics, areas of difficulty, learning goals, academic level, preferred tutor characteristics, learning mode, languages, budget, availability and location. Students can create, edit, activate and remove learning needs according to their current academic requirements. An active learning need is selected as the input to the tutor matching and recommendation process.

This structure allows the recommendation process to be targeted to a specific academic requirement. For example, the same student can maintain separate learning needs for different subjects or academic goals and receive tutor recommendations based on the learning need currently selected. The learning need therefore provides the structured representation of student requirements used by the subsequent matching and recommendation modules.

4.2.3 MULTIMODAL TUTOR VERIFICATION MODULE

Tutor verification is incorporated to improve the reliability and credibility of tutor profiles. The verification workflow combines document-based evidence with available identity-related evidence rather than relying on a single verification method.

For certificate verification, uploaded qualification documents are processed using PaddleOCR to extract relevant information such as the tutor's name, institution, qualification and other available textual fields. The extracted information is compared with the tutor's registered profile using consistency checks. Document security analysis is also performed to identify potential document-level risks and suspicious identifiers. The verification process therefore provides structured evidence regarding the consistency and validity of the submitted qualification information.

When a usable portrait is available on the submitted certificate, facial evidence can also be evaluated. OpenCV-based face detection is used to locate facial regions, while SFace provides a facial representation that can be used to obtain an identity similarity signal between available face images. The current implementation treats this similarity as supporting verification evidence rather than as a standalone automatic identity decision.

The live-face stage additionally uses randomized head-turn instructions and image-based liveness analysis to obtain evidence that the submitted frames originate from a live subject. The liveness mechanism analyses changes between captured frames, while face detection confirms the presence of a detectable face. Certificates without a usable portrait are not automatically rejected; instead, the remaining available verification modalities are used.

Based on the available evidence, the verification workflow can result in a verified, failed or pending state, while cases requiring additional assessment can be routed to the manual verification review module. This multimodal approach allows the system to combine certificate information, profile consistency, document security and available facial and liveness evidence before a tutor becomes eligible for production recommendations.[4], [11].

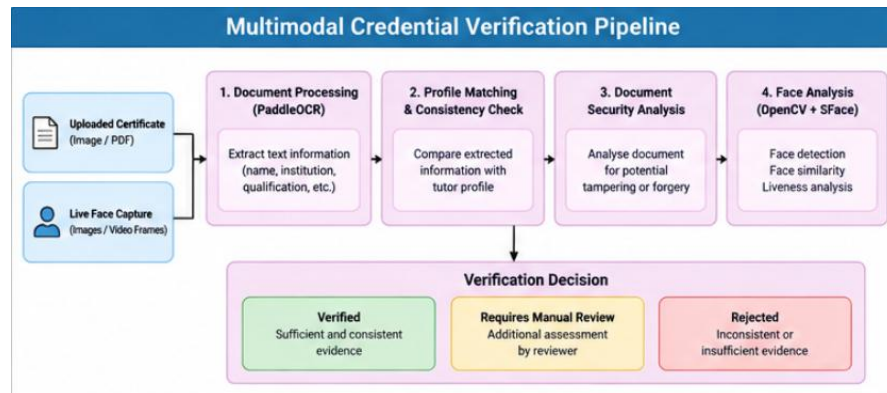


Figure 4.1. Multimodal Credential Verification Pipeline

The verification pipeline combines certificate processing, OCR-based information extraction, profile consistency checking, document security analysis and available facial and liveness evidence. The resulting verification state determines whether the tutor can proceed directly or requires further assessment.

4.2.4 AI-BASED TUTOR MATCHING MODULE

The The AI-based tutor matching module forms the core recommendation component of TutorLinkAI. The module compares a student's selected learning need with the capabilities represented in verified tutor profiles. Instead of relying only on keyword-based filtering, the proposed approach combines semantic similarity, topic compatibility and structured contextual factors to identify tutors whose expertise is relevant to the student's requirements.

Student learning requirements and tutor expertise are represented using sentence embeddings generated through Sentence Transformers. The resulting vector representations allow the system to measure semantic similarity between descriptions even when different terminology is used to express related concepts. In addition to semantic similarity, topic-level compatibility is evaluated by comparing the topics required in the learning need with the topics represented in the tutor's expertise.

The learning need compatibility component is calculated using semantic similarity and topic compatibility. When topic information is available, the learning need score is calculated as:

$$\text{Learning Need Score} = 0.60 \times \text{Semantic Similarity} + 0.40 \times \text{Topic Compatibility}$$

The overall tutor recommendation score then combines learning need compatibility with contextual factors:

$$\text{Overall Score} = 0.45 \times \text{Learning Need} + 0.20 \times \text{Location} + 0.20 \times \text{Fee} + 0.15 \times \text{Time}$$

The location component considers geographical proximity, while the fee component evaluates compatibility between the tutor's pricing and the student's specified budget. The time component considers compatibility between the student's preferred availability and the tutor's available time slots. Teaching mode, language and pedagogical compatibility can also be evaluated when the corresponding information is available.

Only tutors satisfying the required verification eligibility are considered for production recommendations. The resulting scores are used to rank eligible tutors according to their compatibility with the selected learning need and contextual preferences.

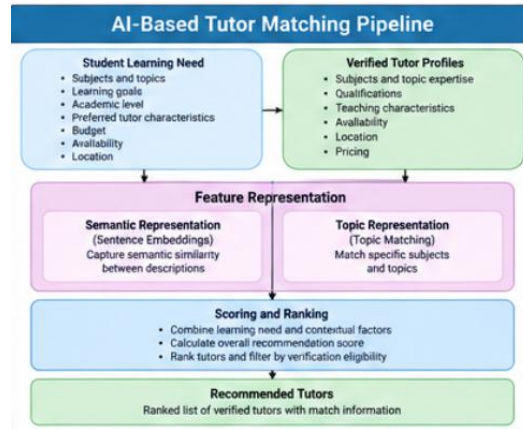


Figure 4.2. AI-Based Tutor Matching Pipeline

The matching pipeline begins with the selected student learning need and verified tutor profiles. Student requirements and tutor expertise are represented using sentence embeddings, while topic compatibility is evaluated alongside contextual factors such as location, fee and availability. The resulting scores are combined to produce a ranked list of eligible tutors. Supporting match information is then provided to the recommendation component for explainable tutor selection.

4.2.5 TUTOR RECOMMENDATION AND BOOKING MODULE

Following the matching stage, the tutor recommendation component presents verified tutors in ranked order according to their compatibility with the student's selected learning need. The recommendation output includes supporting match information so that students can understand the factors contributing to the ranking rather than receiving only an overall recommendation score.

The recommendation information can include matched topics, relevant learning requirements, contextual compatibility and pedagogical compatibility where the corresponding information is available. A score breakdown is also provided to indicate the contribution of major matching factors such as learning need compatibility, location, fee and availability. This provides an interpretable connection between the matching calculations and the tutor recommendations presented to the student.

The booking component extends the recommendation workflow by allowing a student to select a suitable verified tutor and initiate a booking associated with the selected learning

need. The booking workflow maintains the relationship between the student, tutor and selected learning need while recording relevant booking information such as the tutor's applicable rate, booking status and scheduled details. Access to booking operations is controlled according to the user's role and ownership of the booking.

Together, the recommendation and booking components provide an end-to-end workflow from AI-based tutor discovery to tutor selection and booking. This allows the student to review the basis of a recommendation before proceeding with the booking process.

4.3 MATCHING EVALUATION METHODOLOGY

An offline evaluation framework was developed to assess the performance of the tutor matching component under controlled conditions. The evaluation uses a curated benchmark consisting of structured student learning queries, candidate tutor profiles and relevance judgements. The benchmark contains 20 student queries and 30 tutor candidates, producing 600 query-candidate relevance pairs for controlled comparison of different matching approaches.

Four matching configurations are evaluated: a legacy profile-based baseline, dense semantic matching, topic-overlap matching and a full hybrid configuration combining learning-need compatibility with contextual factors. The comparison is intended to examine the contribution of semantic representation, topic-level compatibility and the combined ranking approach.

Ranking performance is evaluated using Precision@1, Precision@3, Precision@5, Recall@5, Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG) and pairwise ranking accuracy. These metrics provide different views of recommendation quality, including the relevance of highly ranked tutors, retrieval of relevant candidates and the ordering of relevant tutors within the recommendation list.

The benchmark is designed for controlled offline evaluation of the matching approaches and does not represent live production user behaviour. The results are therefore used to compare the implemented configurations under the same evaluation conditions rather than to make claims about real-world user performance.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The implementation of TutorLinkAI integrates the major workflows described in the proposed system, including authentication, student learning need management, tutor onboarding and profiling, multimodal tutor verification, AI-based tutor matching, explainable tutor recommendation and booking. The application was developed using a React-based frontend, FastAPI backend services and PostgreSQL for persistent data storage. Separate workflows are provided for students, tutors and verification reviewers, with role-based access controlling the operations available to each user.

The implemented system provides an end-to-end workflow in which students can create and manage learning needs, tutors can complete their profiles and submit qualification documents, and the verification pipeline can process the available credential and identity-related evidence. Verified tutors are then made eligible for the production matching workflow, where the selected student learning need is compared with tutor expertise using semantic similarity, topic compatibility and contextual factors. Ranked recommendations are presented with supporting match information, after which the student can proceed to the booking workflow.

The implementation therefore extends beyond basic tutor registration and profile display by connecting learning need modelling, tutor verification, AI-based matching, explainable recommendation and booking within a single platform.

5.1.1 AUTHENTICATION AND USER MANAGEMENT

The authentication and user management component was implemented to provide controlled access to the TutorLinkAI platform. Separate user roles are maintained for students, tutors and verification reviewers, with role-based access determining the operations available to each user.

The authentication workflow supports user registration, login and protected access to application resources. Backend API endpoints validate the authenticated user's role and ownership before allowing operations on protected data. This prevents users from accessing

learning needs, tutor profiles, verification records or booking information belonging to other users.

The implementation establishes authentication as a common foundation for the remaining modules of the system. Student-specific functions, tutor-specific functions and verification reviewer operations are therefore handled through controlled role-based workflows.

5.1.2 STUDENT LEARNING NEED MANAGEMENT

The student learning need management component allows students to represent different academic requirements separately instead of maintaining a single static requirement profile. Each learning need stores information such as subjects, specific topics, learning goals, academic level, preferred tutor characteristics, budget, availability and location. Students can create, edit, activate and remove learning needs according to their current requirements.

The active learning need is passed to the matching component as the basis for tutor recommendation. This allows the same student to maintain different requirements for different subjects or academic goals and receive recommendations targeted to the learning need currently selected.

5.1.3 TUTOR ONBOARDING AND PROFILING

The tutor onboarding component allows tutors to register and create profiles containing their personal, academic and professional information. Tutor profiles include qualifications, subjects taught, areas of expertise, topic-level expertise, teaching preferences, languages, experience, location, availability, pricing and verification status.

The implementation separates tutor registration from qualification verification. This ensures that creating a tutor account does not automatically establish the tutor as verified. The profile information collected during onboarding is subsequently used as structured input for the verification and matching workflows.

5.1.4 MULTIMODAL TUTOR VERIFICATION

The uploaded qualification certificates are processed using PaddleOCR to extract relevant information and compare it with the tutor's registered profile. Document security analysis identifies potential document-level risks, while OpenCV face detection, SFace facial

representation and live-face liveness analysis provide additional identity-related evidence when available. Certificates without usable portraits are processed using the remaining verification modalities, with cases requiring further assessment routed for review.

5.1.5 AI-BASED TUTOR MATCHING

The matching component compares the selected learning need with verified tutor profiles using sentence embeddings, topic compatibility and contextual factors. The ranking incorporates learning need compatibility, geographical proximity, fee compatibility and availability, while only eligible verified tutors are considered for production recommendations.

5.1.6 EXPLAINABLE TUTOR RECOMMENDATION AND BOOKING

Ranked recommendations are presented with supporting information such as matched topics, contextual compatibility and score contributions. Students can select a verified tutor and initiate a booking associated with the selected learning need, providing an integrated workflow from recommendation to booking.

5.1.7 MATCHING EVALUATION RESULTS

The matching component was evaluated using an offline benchmark containing 20 student queries, 30 tutor candidates and 600 query-candidate relevance pairs. Four configurations were compared: a legacy profile-based baseline, dense semantic matching, topic-overlap matching and the full hybrid approach. Evaluation used Precision@1/3/5, Recall@5, MRR, NDCG and pairwise ranking accuracy. The benchmark provides a controlled comparison of matching approaches and does not represent live production user behaviour.

5.2 DISCUSSION

The implementation integrates tutor verification with learning-need-aware recommendation rather than relying only on tutor profiles or proximity. The matching component combines semantic, topic and contextual factors, while explainable recommendations provide supporting information for tutor selection.

The system has limitations: OCR performance depends on document quality, identity-related analysis depends on the available evidence, and the verification workflow cannot independently confirm the legitimacy of an issuing institution without external institutional data. The offline matching benchmark also does not represent live user behaviour.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

TutorLinkAI addresses the problem of determining whether a tutor is genuinely qualified. The platform does not leave this question to word of mouth; it combines formal onboarding with automated checks on credentials and provides tutors with a profile against which they can be assessed. Parents obtain a tutor profile with a verification status attached, rather than only a name and phone number passed on by a relative, which turns the selection of a tutor into an assessment that a parent can make.

For tutors, the platform provides a clear way to prove their qualifications, in place of a resume that is not independently checked. The current version shows that credential verification can be added to a tutoring platform and can function without every document being checked manually, giving parents verifiable information in place of a suggestion from an acquaintance.

6.2 FUTURE WORK

The current implementation provides a foundation for automated tutor verification and recommendation. Further work can build upon this framework by improving credential verification, student performance analysis, security and personalised learning support. The following extensions are identified:

7. **Institutional credential verification:** integration with university, educational board and certification databases to verify tutor credentials directly from authorised sources, improving the reliability of verification and reducing invalid submissions [4], [11].
8. **Advanced document fraud detection:** image-forensics and machine-learning techniques to identify forged, manipulated or digitally altered documents, adding a layer of security to tutor verification [3].
9. **Early student performance prediction:** predictive machine-learning models to analyse learning patterns, assessment results and interaction data and identify

students at risk of poor performance, so that tutors can provide timely feedback and targeted support [1], [9].

10. **Feedback and rating analysis:** Natural Language Processing (NLP) techniques to analyse student and parent feedback on tutoring sessions and identify common patterns in tutor performance, supporting more informed choices.
11. **Continuous tutor verification:** periodic reassessment of tutor credentials and profile information so that verification remains current.
12. **Scalable cloud deployment:** cloud infrastructure and containerisation to support an increasing number of students, tutors and concurrent verification requests, improving scalability and availability.
13. **Enhanced security and privacy:** stronger encryption, role-based access control, audit logging and privacy-preserving storage to protect sensitive credentials and academic documents [5], [12].
14. **Multimodal AI-based doubt solving:** a multimodal AI module capable of processing text, images and other resources to provide learning aid and resolve academic doubts, offering interactive and context-aware learning support alongside tutor recommendations.

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TUTORLINKAI: A Verified Local Tutor Discovery and Booking Platform

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Abstract- In the fast-paced world, education plays a vital role in shaping the future of the younger generation. Many students find it difficult to keep up with their classes and compete academically with their peers. This creates a need for additional academic support for the students, leading to parents seeking tutors who can help their children understand the concepts where they require more assistance.

Parents often choose tutors who are close to their homes or tutors who have been suggested by relatives or neighbours;

they miss out on checking if the tutor is the right fit for their child.

It is given that the parents want what is best for their child and help them academically, but they do not always perform background checks and assess whether this tutor is appropriate for the needs of their child and if they are properly qualified.

This often leads to the child not being able to understand the concepts despite the parents providing them with some help, but just not the right resources.

Another major problem that parents and students face is the lack of verification for the tutors. We must understand who is helping our children and assess if that person is legitimate and trustworthy. It is also important that we perform a credibility check rather than blindly believing the neighbors' or relatives' suggestions.

This is where TutorLinkAI comes in; the main goal of this application is to provide parents with the right information

to help provide the children with the right resources that help them academically without adding any strain.

TutorLinkAI focuses on connecting parents with the right tutors by assessing their qualifications and seeing if they can help in subjects where the child needs help.

Using TutorLinkAI, parents can check if the tutor is actually credible and has the appropriate degrees and credentials to help their children; this creates a sense of relief for both the children and the parents.

Keywords: TutorLinkAI, Online Tutoring, Tutor Verification, Credential Verification, Optical Character Recognition (OCR), Artificial Intelligence, Tutor Recommendation, Educational Technology, Academic Support, Trust and Safety

I. INTRODUCTION

Learning has increasingly moved beyond the confines of a traditional classroom. Given the academic background of students and their level of comprehension, classroom teaching may not always be enough and as such tuition services have long provided additional support to students who require more help in certain areas of learning than what is provided in the classroom. However, the efficiency of

such assistance depends on locating a tutor who is qualified, experienced, and able to provide guidance to the student according to their specific needs.

Traditionally, students and parents have relied on a number of private networks, including family, friends, neighbours in order to locate tutors. While such methods can be effective in shortlisting potential tutors, the information provided is usually limited. Factors to consider when selecting a tutor include qualifications, the subject matter or area of expertise that the student needs help with, experience, location, availability, and fees amongst other considerations. Gathering and comparing this information can be a tedious process which many parents and students find inconvenient.

Online services have been developed to enable users to first research and then locate a suitable service based on certain criteria. However, there are various limitations associated with existing tutor discovery methods and services. Firstly, many existing platforms do not allow parents or students to easily locate tutors using certain criteria, as well as comparing tutors based on certain requirements. Such an option can prove especially useful to parents and students looking for specific tutors. In many cases, finding a tutor relies heavily on word of mouth, which limits students and parents to tutors within their personal networks and provides limited opportunities to match tutors effectively with the specific needs and requirements of the student.

Even further into the tutor discovery process, parents and students must engage in separate communication channels when contacting a tutor. As such, parents and students must individually contact a tutor in order to confirm their qualifications, availability, as well as schedule and reschedule sessions. This can be inconvenient and in many cases redundant. A proper platform can enable both students and parents to communicate with tutors and manage scheduling and other activities related to the tutoring sessions.

An equally important limitation in the tutor discovery process is the reliability of information provided to parents and students. It is vital that both students and parents consider a number of factors when selecting a tutor, including educational background, experience, and the area of expertise. Making such information easily accessible to parents and students can be extremely valuable in ensuring that both parties have realistic expectations of the tutor's qualifications and ability.

The proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at addressing the various limitations of the traditional methods of tutor discovery. By aggregating tutor related information and offering a wide range of features for both parents and tutors, TutorLinkAI facilitates

the tutor discovery process based on a number of criteria including subject, location, qualifications, availability, experience and affordability. By providing tutors with the ability to post their information and experience for students and parents to view, the proposed solution addresses the limitations of relying on word of mouth by offering both parents and students a wider pool of tutors to consider. Additionally, the proposed platform enables students and parents to sort through the available tutors according to their specific needs, thus eliminating the need to manually sift through hundreds of tutors or advertisements.

The proposed Local Tuition Discovery Platform, TutorLinkAI, supports the United Nations Sustainable Development Goal (SDG) 4: Quality Education by enabling students to receive more personalized supplementary education. By enabling proper matching of students and tutors based on student requirements as well as tutor qualifications, which focus on enhancing personalized learning and promoting access to quality education for all.

Apart from supporting universal access to quality education, TutorLinkAI aligns with SDG 10: Reduced Inequalities, which emphasizes the need for empowering economically disadvantaged individuals. By providing affordable means of accessing supplementary education and offering students and parents the option of sorting tutors based on their budget, TutorLinkAI can significantly help reduce the disparities associated with access to additional academic services.

As such, the proposed Local Tuition Discovery Platform, TutorLinkAI, is aimed at enabling personalized education by addressing the inefficiencies associated with accessing information regarding local tuition services. By incorporating local tuition services, offering additional information on tutors' qualifications and experience, as well as providing various tools for filtering and matching tutors with students based on their specific needs and budgets, TutorLinkAI facilitates more convenience and transparency in locating potential tutors.

II. LITERATURE SURVEY

The literature survey provides a theoretical framework and identifies prior research relevant to the design and development of intelligent online tutor discovery and matching services. Existing solutions primarily aim to bridge the gap between students and suitable tutors through subject-based searches, location-based discovery and booking facilities. The early approaches depend predominantly on the profile information provided by the user and predefined search categories, providing limited support for personalised and student-centric tutor recommendations [6]-[8].

Several online tutor discovery systems have investigated structured student-tutor management and filtering mechanisms. These applications have incorporated criteria such as education level, subject expertise and geographical location for tutor filtering. A comparative analysis of existing applications student and tutor registration and tutor filtering are common functionalities, while features such as tutor-parent interaction and communication, tutor ranking provide additional support. These approaches demonstrate the feasibility of recommendation-based tutor discovery while also indicating the importance of incorporating multiple learner and tutor characteristics [2].

Location-based approaches have improved tutor discovery by incorporating geographical context such as latitude and longitude information to map tutor and student locations and filter tutors based on proximity. This is particularly practical in cases where travelling distance is a priority. Beyond online tutor-finding applications, online location-based service recommendation systems provide advanced mechanisms to combine proximity with other attributes. These systems employ mechanisms such as the Haversine formula to calculate geographical distance and use K-Nearest Neighbours (KNN) with scoring mechanisms based on factors such as proximity, price and ratings to rank service providers. These systems support service provider availability, booking and reviews. Such approaches demonstrate that geographical proximity can be incorporated with multiple other factors to determine the most suitable service provider. However, proximity alone is not sufficient to determine academic suitability and can be combined with other features, such as subject expertise, tutor reviews, and teaching approach, to identify a suitable tutor for a student's particular learning needs.

The limitations of static filtering have led to the development of personalized recommendation systems that incorporate user preferences along with behavioral information and service provider characteristics to generate relevant recommendations[6], [14]. Machine learning based recommendation systems use historical data such as user-provider relationships and combine it with contextual re-ranking and factors such as urgency, provider availability, ratings and user interactions. This demonstrates how such systems can move beyond simple filtering by learning user-provider relationships. For the tutor recommendation system the same principle can be used by considering students and parents as users and tutors as education providers. Additional features can be incorporated, including subject and topic expertise, academic level, student skill requirements, teaching preferences, and learning objectives.

An important extension to conventional tutor recommendation systems is the incorporation of student skill and knowledge assessment. A student may identify a subject as a requirement without being able to accurately identify the specific concepts in which additional assistance is required. Knowledge-tracing techniques provide a mechanism for addressing this issue by estimating a learner's evolving knowledge state from historical learning interactions [1]. One such approach is a Transformer-based Deep Knowledge Tracing framework for formative assessment of student skills [9]. The framework analyses sequences of student interactions to estimate knowledge states and identify knowledge gaps. The resulting learner representation can provide valuable information for tutor recommendation because it can distinguish between a student needing assistance across the entire subject and a student who specifically requires help in specific topics. Although knowledge-tracing research primarily focuses on student assessment, its resulting learner representation can be used as an input to identify suitable tutors.

Technical approaches such as semantic representation of textual information provide another means of improving recommendation systems beyond keyword matching. Sentence embedding models represent textual information in vector spaces where semantic relationships are measured computationally. Ni et al. proposed Sentence- T5 for scalable sentence encoding and evaluated its effectiveness for semantic textual similarity tasks [10]. This approach provides a basis for comparing descriptive student requirements with tutor expertise, particularly when different terms are used to express related concepts.

Automated document analysis can further support credential verification by identifying potentially manipulated documents. Al-Ameri et al. investigated an unsupervised method for detecting document forgery using document analysis techniques [3]. Such approaches demonstrate that document-level analysis can complement text extraction and profile consistency checks. In the current TutorLinkAI implementation, OCR is primarily used to extract information from the submitted certificates, while advanced document forgery detection is a potential future extension.

The literature therefore indicates that tutor discovery can be viewed as a multi-criteria recommendation problem. Educational compatibility can be determined from factors such as student skill requirements, subject or topic preferences and tutor skills and expertise, while contextual feasibility can be incorporated through factors such as pricing, availability, distance, ratings and tutor profile verification. Existing tutor recommendation systems demonstrate the importance of profile-based matching, location-based systems demonstrate the usefulness of

geographical and service constraints, machine-learning-based recommendation systems highlight the importance of behavioural and contextual personalization, and knowledge-tracing research highlights the importance of student assessment for identifying knowledge levels and mapping student needs. However, these capabilities are generally addressed as separate problems rather than being integrated into a unified tutor recommendation process.

III. EXISTING SYSTEM AND PROBLEM IDENTIFICATION

Existing solutions are mostly old-fashioned coaching institutes where their main goal is to increase student enrollment every year. This causes a financial drain on families and prevents students from finding the right help. These coaching institutes often take classes for 20-30 students at once, leading to a lack of personalisation and failing to cater to the individual needs of each student.

Most centers also assign the students homework assignments, adding to the physical and mental exhaustion of the students, which in turn again leads to them not performing as expected academically.

Although one can argue the demerits of the existing traditional methods, it is also important that we recognise the few centers that actually upskill the students and help them improve their academic capabilities through their extensive coaching.

Some existing online tutor finder platforms map the students to tutors who reside in a different pincode; this adds to the physical exhaustion of the students.

The platforms often advertise their fancy courses, which usually end up as clickbait for increased enrollments; additionally, students having to choose from a wide variety of courses and faculties can become hectic.

Students usually have limited information about the availability of tutors in their localities, tutors who can help them in the subjects they are struggling with, and help them find tutors who charge fees that are economical for them.

Major Drawbacks of Existing Solutions

There are several ways for students to find tutors, but most of the existing methods still have some drawbacks.

1. Commercial Coaching Institutes and Manual Tuition Centers

In most cases, students find local tuition centers or coaching classes through advertisements, recommendations, or by asking around. Often the classes can end up being expensive, or they can be taught in groups, hence leaving the tutor to not be able to cater to individual student needs.

Finding the right tutor can also take time because parents often have to make calls or visit different centres.

2. Web and Directory-Based Tutor Portals

Examples: A+ Home Tuition, bdtutors.com and similar websites.

Online websites let students search for tutors using basic details like subject and location, and do not focus on important factors like student needs, affordability, and class scheduling. These platforms do not have direct communication between the tutors and the parents, and hence arranging classes may take extra time. The matching is also mostly based on profile details rather than what the student actually needs help with.

3. Multi-Criteria and Location-Based Platforms

Examples: GRAB Tutor Management System (GTMS), Tutors Map and Tutor Finder (LBS).

These platforms allow students to search for tutors using filters based on subject, gender, and expertise. They mainly use the information given in the tutor's profile. These platforms do not consider factors such as students' weak areas, teaching quality, availability, and fees.

Overall, most existing solutions focus on finding tutors using basic information or past activity. They do not give enough importance to the student's actual learning difficulties, current performance, tutor availability, and cost.

IV. PROPOSED METHODOLOGY

The application TutorLinkAI is a secure and intelligent platform that is designed to help parents and students in identifying the suitable and credible tutors. The application addresses two of the major limitations in the existing tutor selection system, i.e., the difficulty of assessing the tutor suitability and the lack of reliable tutor verification in the existing systems.

The proposed system consists of a tutor onboarding module, credential verification module, tutor profile management module, and tutor discovery and matching module. During onboarding, tutors register on the platform by providing their personal, academic, and professional information. An authentication mechanism is used to verify the tutor's account before the profile is created.

A key component of TutorLinkAI is the tutor credential verification process. Tutors are required to upload relevant qualification documents as part of the onboarding process. The uploaded documents are processed using Optical Character Recognition (OCR) to extract relevant textual

information. The extracted information is then analyzed and compared with the information provided by the tutor during registration. Similarity-based validation is used to identify inconsistencies between the submitted credentials and the registered profile information. Based on the verification results, the system identifies whether the submitted information is consistent with the registered tutor profile and assigns an appropriate verification status.

The backend of the application is implemented using Python and FastAPI, this provides RESTful APIs for authentication, onboarding, profile management, document processing and verification. PostgreSQL is used for persistent data storage, with SQLAlchemy providing database interaction and Alembic supporting database schema migration. The frontend provides interfaces for tutor registration, profile creation, credential submission, and tutor discovery.

In the application, once the tutors are verified, the students and parents can view their profiles and identify tutors based on relevant academic requirements. The application combines identity onboarding, credential verification and tutor suitability assessment within one platform, hence providing a structured alternative to relying solely on recommendations from relatives, neighbors or nearby tutors.

V. SYSTEM DESIGN AND IMPLEMENTATION

The proposed TutorLinkAI system is designed as a modular intelligent platform for tutor discovery and personalized tutor recommendation. The system integrates student and tutor profiling, tutor verification, student requirement assessment, AI-based matching, and booking within a unified platform. The architecture separates the user interface, application logic, data management, and AI-based processing components to support modular development and future extension.

A. System Architecture

TutorLinkAI follows a modular client-server architecture consisting of a React-based frontend, FastAPI backend, PostgreSQL database and dedicated AI services. The frontend provides interfaces for student and tutor registration, profile management, assessment, tutor discovery, recommendation, and booking. The backend manages authentication, application logic, database operations and communication between the frontend and AI services. AI services handle computationally intensive operations such as semantic matching, face verification and certificate OCR.

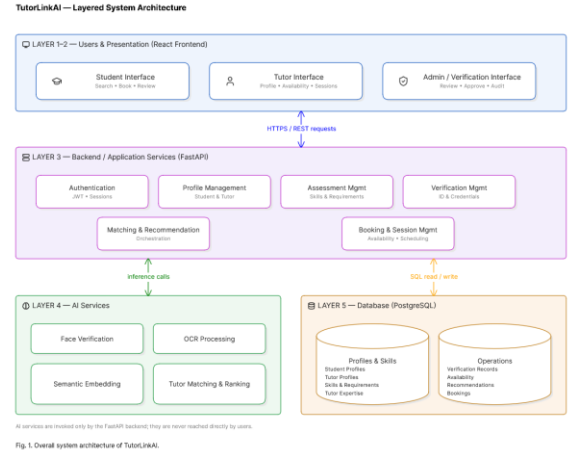


Fig. 1. Overall system architecture of TutorLinkAI.

B. Student and Tutor Profiling

The system provides separate processes for registration and profile management for students and tutors. The student profile captures information related to their academic requirements, subject skills, learning objectives, location, availability, and preferred budget. Tutor profiles contain information including subjects taught, areas of expertise, qualifications, experience, location, availability, pricing and verification status.

The collected information then forms the basis for subsequent assessment, verification, matching and recommendation processes. This information constructs the overall profile of the user.

1) Student Requirement Profile

The student's learning requirements are collected during the initial stage in the form of student registration and initial assessment. Rather than relying on selected subjects the system captures specific topics, areas of difficulty, skill level and learning objectives. This enables the matching component to identify the most suitable tutor according to the student's preferences and requirements.

2) Tutor Expertise Profile

Tutor expertise is represented using information such as subjects, topics, qualifications and teaching expertise. Both structured attributes and descriptive expertise information can be utilized during the matching process.

C. Tutor Verification

Tutor verification is incorporated to improve the reliability and credibility of tutor profiles. The verification module combines facial verification and document-based verification.

1) Face Verification

The system processes the tutor's submitted facial image to obtain a facial representation for identity comparison. OpenCV can be used for image processing, with InsightFace which is used for feature detection and facial structure detection. The resulting similarity information is used as one of the signals in determining the tutor's verification status [5], [12].

2) Certificate Verification

Uploaded qualification certificates are processed using Optical Character Recognition (OCR). PaddleOCR is used to extract relevant information from certificates such as the tutor's name, institution, qualification and other available textual fields. OCR-based extraction provides structured information that is compared against the tutor's profile information with stronger institutional validation as a future extension [4], [11].

D. AI Based Tutor Matching

The core component of TutorLinkAI is the tutor matching and recommendation system. Instead of relying solely on keyword-based or predefined filtering, the proposed approach considers both semantic similarity and structured compatibility between the student and tutor.

Student requirements and tutor expertise may be represented using sentence embeddings generated through Sentence Transformers[10], [13]. Semantic similarity between these representations can then be used to identify tutors whose expertise is relevant to the student's requirements.

E. Tutor Recommendation and Booking

Following the matching stage, suitable tutors are ordered according to how closely they satisfy the student's needs. Through the recommendation portal, students can review detailed tutor profiles covering qualifications, subject expertise, identity verification, location, pricing, and schedule to help them choose the most appropriate match.

The booking component maintains information regarding selected tutors, students, session schedules, and booking status. This provides an end-to-end workflow from tutor discovery to session scheduling.

VI. RESULTS AND DISCUSSIONS

The implementation of TutorLinkAI shows that it is possible to combine tutor onboarding and credential verification into a tutoring platform. The system that was built offers a step-by-step route for tutors to sign up, submit their qualifications and get checked before they appear as possible tutoring resources.

The verification module is where the most beneficial occurs. The system pulls out the text and examines them in, yet compared to what the tutor documented when they signed up. Very soon after we started developing it, we realized one thing: you can't have registration and verification happen at the same time. The tutor account does not prove the degree behind it is real. Splitting these into separate stages gives the platform a real basis for calling someone "verified" rather than just "signed up."

It's a major shift for parents in terms of decision making. They are not picking a tutor because a neighbor mentioned a name or someone lives two streets down the road. They can balance qualifications and verification status with location, which is a very different, and arguably more responsible, way to choose who teaches your kid.

But it is not without its flaws, OCR is only as good as the document that goes into it, and a blurry photo, bad handwriting or unusual certificate format can throw off the accuracy of extraction. The similarity check has a similar ceiling, it can tell you the extracted text doesn't match the registration form, but it has no way of confirming that the issuing institution itself is legitimate unless there's an external database to check against. So for now, the verification layer is an automated first pass at credibility not a replacement for an actual institutional check.

VII. CONCLUSION

TutorLinkAI solves a problem that has been poorly solved for a long time: how do you know if the tutor you are about to hire really knows what they say they know? The platform doesn't leave that question to word of mouth, it combines formal onboarding with automated checks on credentials and provides tutors with a real profile to stand behind.

Parents get something they never had before: a profile of the tutor, with actual verification status attached, not just a name and phone number passed along by a relative. This is the shift TutorLinkAI aims to achieve. Taking the guesswork out of choosing a tutor and turning it into an assessment a parent can make.

For tutors, the platform offers a real advantage. A tutor who has qualifications now has a clear way to prove those qualifications, not just a resume that nobody checks. Over time this will help separate qualified educators from people who are only good at self-promotion.

What this version shows is that it is possible to add credential verification to a tutoring platform that it can work without a person checking every paper by hand and that it gives

parents something real to see instead of a suggestion, from someone living three houses away. That is the gap TutorLinkAI aimed to fix and in that way it does what it was supposed to do.

VIII. FUTURE SCOPE

The current implementation of TutorLinkAI provides a foundation for automated tutor verification and recommendation. Further work can build upon this framework by improving credential verification, student performance analysis, security and personalized learning support.

1. Institutional Credential Verification: The system can be integrated with university, educational board and certification databases to verify tutor credentials directly from authorized sources. This mechanism can improve the reliability of the verification process and reduce submissions of invalid credentials and information[4], [11].
2. Advanced Document Fraud Detection: Image-forensics and machine-learning techniques can be incorporated to identify forged, manipulated or digitally altered documents by providing an additional layer of security in the tutor verification process[3].
3. Early Student Performance Prediction: Predictive machine learning models can be incorporated to analyze student learning patterns, assessment results and interaction data to identify students who may be at risk for poor performance. These early detection mechanisms can be helpful for tutors to provide the student with timely feedback and targeted support[1], [9].
4. Feedback and Rating Analysis: Natural Language Processing (NLP) techniques

can be used to analyse the student and parent feedback to tutoring sessions and identify common patterns in tutor performance. These extracted insights support students in making more informed decisions while choosing their tutors.

5. Continuous Tutor Verification: The platform can periodically reassess tutor credentials and profile information to ensure that verification remains current.
6. Scalable Cloud Deployment: Cloud infrastructure and containerization can be adopted to support an increasing number of students, tutors and concurrent verification requests. This improves the scalability and availability of the system.
7. Enhanced Security and Privacy: Future implementations can incorporate stronger encryption, role-based access control, audit logging, and privacy preserving storage mechanisms to protect sensitive credentials and academic documents[5], [12].
8. Multimodal AI-Based Doubt Solving: The platform can be extended with a multimodal AI module which is capable of processing text, images and other resources for providing the students with learning aid and resolving academic doubts. This system can provide interactive and context-aware learning support alongside tutor recommendations.

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